Michal Malyska

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Work Experience

Machine Learning Operations Lead — **Toronto, Canada**
Fable Therapeutics — *January 2024 - Present*

- Reduced training compute from ~1,920 to ~80 H100 GPU-hours and inference cost by >99% (~$1 to <$0.01 per prediction) across >90% of production workloads.
- Cut biweekly pipeline cost by ~60% per run ($130k–$170k to $50k–$60k) through compute routing, spot strategy, and model/runtime optimization.
- Re-architected a 27-step design-generation workflow from manually chained notebooks into a config-driven Python orchestration system on AWS Batch, reducing manual interventions by ~70% YoY.
- Reduced per-run operator time from ~2 weeks to ~8 hours by automating handoffs, monitoring, and alerting across compute environments.
- Built CI/CD for model and code pipelines with GitHub Actions and Docker release automation; integrated W&B artifact lineage for reproducible model/config releases.
- Built a bidirectional wet-lab/ML data platform that replaced CSV/email handoffs with a governed Postgres workflow for design delivery and assay-result feedback into model development.

Machine Learning Lead — **Toronto, Canada**
Semantic Health (acquired) — *January 2021 - December 2022*

- Built active-learning and feedback pipelines that reduced labeling cost by >75% and improved downstream KPIs by >50%.
- Created and shipped new ML product lines from customer calls and internal QA findings, adding ~$100k ARR per contract.
- Led development and production delivery of clinical NLP/OCR/tabular ML systems across enterprise deployments, supporting US expansion and a $60M valuation outcome.
- Led MLOps modernization (Feast + MLflow + Seldon) for model versioning, monitoring, and end-to-end release testing in cloud dev/staging environments.
- Rewrote the Python ML stack and standardized code, data, and testing workflows to reduce technical debt and improve onboarding speed.
- Created model and prediction fallback paths that removed the need for ML on-call support in customer-facing operations.

Machine Learning Scientist — **Toronto, Canada**
Semantic Health (acquired) — *June 2019 - December 2020*

- Built evidence-generation and ontology-graph label-smoothing workflows that improved model performance by 35% and product KPIs by 300%.
- Developed transformer-based models that exceeded prior baselines on a large, noisy multi-label clinical dataset used in production.
- Built a multi-armed-bandit model release strategy that increased deployment cadence 4x (monthly to weekly).
- First technical hire; built the initial ML stack and moved deployments from scripts/database uploads to orchestrated pipelines used in long-term customer contracts.

Sessional Lecturer — **Toronto, Canada**
University of Toronto — *2020 - 2023*

- Taught undergraduate and graduate Statistics/ML courses; updated CSC412 (Probabilistic ML) curriculum to include attention, transformers, and diffusion models.

Education

University of Toronto — **2022, 2019**
MSc. Statistics, BSc. Statistics

Publications

- BioNLP - ICDBigBird: A Contextual Embedding Model for ICD Code Classification
- ACM CHIL - Active learning for medical code assignment
- EMNLP - Multiple Sclerosis Severity Classification From Clinical Text

Skills

- **Languages**: Python, TypeScript, SQL, R, JavaScript
- **Tech Stack & ML**: Git, MLflow, Kubernetes, Docker, Dagster, Airflow, Feast, Seldon, ArgoCD, Terraform, PyTorch, TensorFlow, NLP, LLMs